

An Exact Higher-Order Jet Functional on the Real Line

Candidate partial result for arXiv:1211.5700

11 August 2026

Status: candidate partial result, likely valid. For every order $m \geq 1$, an explicit two-jet functional is constructed for scalar m -fields on $\mathbb{R}$. It gives the exact minimum $\text{Lip}(F^{(m)})$, admits an exact extension for arbitrary data sets, and satisfies the source paper's properties (P_0) – (P_5) . The problem in dimensions $d \geq 2$ remains open.

1 Source question

For 1-fields on $\mathbb{R}^d$, Le Gruyer's functional Γ^1 is the exact minimum Lipschitz constant of the gradient and satisfies the six structural properties used by Hirn and Le Gruyer. Section 2.4.5 of their paper then states [1]:

“A similar result for m -fields, where $m \geq 2$, is an open problem. In particular, it is unknown what the correct corresponding functional $\Phi = \Gamma^m$ is. It seems plausible, though, that such a functional will satisfy the properties (P_0) – (P_5) .”

10

Matthew J. Hirn, Erwan Y. Le Gruyer

1. $J_x F = f(x)$ for all $x \in E$.
2. $\text{Lip}(\nabla F) = \Gamma^1(f; E)$.
3. If $\tilde{F} \in C^{1,1}(\mathbb{R}^d)$ also satisfies $J_x \tilde{F} = f(x)$ for all $x \in E$, then $\text{Lip}(\nabla F) \leq \text{Lip}(\nabla \tilde{F})$.

Thus F is the extension of the 1-field f with minimum Lipschitz derivative (see [18] for the proofs and a complete explanation). The 1-field U is the corresponding set of jets of F . For an explicit construction of F when E is finite we refer the reader to [20].

2.4.5. m -fields A similar result for m -fields, where $m \geq 2$, is an open problem. In particular, it is unknown what the correct corresponding functional $\Phi = \Gamma^m$ is. It seems plausible, though, that such a functional will satisfy the properties (P_0) – (P_5) .

2.5. Main theorem

The AMLE condition (7) is for any open set off of the domain of the initial function f . In our analysis, we look at subfamily of open sets that approximates the family of all open sets. In particular, we look at finite unions of open balls.

Figure 1: The m -field open problem in arXiv:1211.5700v3, PDF page 10, Section 2.4.5.

This packet resolves that functional problem on the one-dimensional domain $\mathbb{R}$, for all orders and arbitrary (not necessarily closed or finite) data sets. It does not claim a solution in higher-dimensional domains.

2 Definitions and the proposed functional

Let $\mathcal{P}_m$ be the real polynomials of degree at most m . An m -field on $E \subset \mathbb{R}$ is a map $f : E \rightarrow \mathcal{P}_m$. Write the jet at x as

$$f(x)(a) = \sum_{k=0}^m \frac{p_k(x)}{k!} (a-x)^k, \quad p_k(x) = f(x)^{(k)}(x).$$

For $F \in C^{m,1}(I)$, $J_x^m F$ denotes its degree- m Taylor polynomial at x , and

$$\text{Lip}_I(F^{(m)}) = \sup_{s \neq t \in I} \frac{|F^{(m)}(s) - F^{(m)}(t)|}{|s - t|}.$$

For $x < y$, $P, Q \in \mathcal{P}_m$, define the two-jet cost

$$\Gamma_m(P, Q; x, y) = \min \left\{ \text{Lip}_{[x,y]}(F^{(m)}) : \begin{array}{l} F \in C^{m,1}([x, y]), \\ J_x^m F = P, J_y^m F = Q \end{array} \right\}. \quad (1)$$

For $x > y$ use the symmetric definition, and set $\Gamma_m(P, P; x, x) = 0$. For a field f put

$$\Gamma_m(f; x, y) = \Gamma_m(f(x), f(y); x, y), \quad \Gamma_m(f; E) = \sup_{x \neq y \in E} \Gamma_m(f; x, y),$$

with the empty supremum equal to zero.

The word “minimum” in (1) is justified below. More importantly, the definition has an explicit finite-dimensional dual formula.

Proposition 1 (Moment and dual formulas). *Fix $x < y$, set $h = y - x$, and write*

$$a_k = P^{(k)}(x), \quad b_k = Q^{(k)}(y), \quad 0 \leq k \leq m.$$

For $0 \leq r \leq m$, define the endpoint discrepancy

$$d_r = b_{m-r} - \sum_{j=0}^r \frac{a_{m-r+j} h^j}{j!}. \quad (2)$$

Then

$$\Gamma_m(P, Q; x, y) = \min \left\{ \|w\|_{L^\infty(0,h)} : d_r = \int_0^h \frac{(h-s)^r}{r!} w(s) ds, \ 0 \leq r \leq m \right\} \quad (3)$$

$$= \sup_{\lambda \in \mathbb{R}^{m+1} \setminus \{0\}} \frac{|\sum_{r=0}^m \lambda_r d_r|}{\int_0^h \left| \sum_{r=0}^m \lambda_r \frac{(h-s)^r}{r!} \right| ds}. \quad (4)$$

Both extrema are attained.

Proof. If F is feasible in (1), its Lipschitz function $F^{(m)}$ has an almost-everywhere derivative $w(s) = F^{(m+1)}(x+s) \in L^\infty(0, h)$, and $\text{Lip}(F^{(m)}) = \|w\|_\infty$. Repeated integration from x gives, for $0 \leq r \leq m$,

$$F^{(m-r)}(y) = \sum_{j=0}^r \frac{a_{m-r+j} h^j}{j!} + \int_0^h \frac{(h-s)^r}{r!} w(s) ds.$$

Thus the endpoint conditions are exactly the moment equations in (3). Conversely, integrating any feasible w reconstructs a feasible F with the same norm. The moment map

$$T : L^\infty(0, h) \longrightarrow \mathbb{R}^{m+1}, \quad (Tw)_r = \int_0^h \frac{(h-s)^r}{r!} w(s) ds,$$

is onto: its adjoint is injective because a polynomial that vanishes almost everywhere on an interval is zero. Hence all two-jet data are feasible.

Formula (4) is the standard dual formula for the quotient norm $\inf_{Tw=d} \|w\|_\infty$, since

$$T^* \lambda(s) = \sum_{r=0}^m \lambda_r \frac{(h-s)^r}{r!} \quad \text{and} \quad \|T^* \lambda\|_{L^1} = \int_0^h |T^* \lambda(s)| ds.$$

Weak-* compactness of bounded subsets of L^∞ and weak-* continuity of T give a primal minimizer. The dual supremum is attained after normalizing its denominator, because the dual space is finite-dimensional. $\square$

3 Proof intuition

On a line, removing a closed data set leaves disjoint open intervals. The two endpoint jets on each bounded gap impose exactly the moment constraints in Proposition 1; a norm-minimizing control $w = F^{(m+1)}$ fills that gap. Every gap can be filled independently because neighboring pieces already agree through order m at their common endpoint. The Lipschitz constant of $F^{(m)}$ is therefore the maximum of the local costs, with no loss.

The only issue for arbitrary data sets is compatibility at accumulation points. A two-point interpolant of cost K yields sharp Taylor remainders of order $K|x-y|^{m+1-k}$. Those estimates force every prescribed derivative value to extend continuously to the closure. They also make the gapwise pasting C^m at every accumulation point. This ordered-gap mechanism is specific to one dimension and is exactly what is unavailable in $\mathbb{R}^d$, $d \geq 2$.

4 Exact extension on the real line

Theorem 2 (Exact all-orders extension on $\mathbb{R}$). *Let $m \geq 1$, let $E \subset \mathbb{R}$, and let $f : E \rightarrow \mathcal{P}_m$ be an arbitrary m -field with*

$$K := \Gamma_m(f; E) < \infty.$$

Then there is $F \in C^{m,1}(\mathbb{R})$ such that

$$J_x^m F = f(x) \quad (x \in E), \quad \text{Lip}_{\mathbb{R}}(F^{(m)}) = K. \quad (5)$$

Consequently, for every D with $E \subset D \subset \mathbb{R}$, the field $U(z) = J_z^m F$ on D is an exact minimal extension:

$$\Gamma_m(U; D) = \Gamma_m(f; E).$$

In particular,

$$\Gamma_m(f; E) = \min\{\text{Lip}_{\mathbb{R}}(G^{(m)}) : G \in C^{m,1}(\mathbb{R}), J_x^m G = f(x) \ (x \in E)\}. \quad (6)$$

Proof. We may assume that E has at least two points; the empty and one-point cases are obtained from a polynomial of degree at most m .

Step 1: sharp Whitney estimates. For $x < y$ in E , choose a minimizer in (1). Repeated integration, now separating the constant value at x from the variation of the m th derivative, gives for every $0 \leq k \leq m$

$$\left| p_k(y) - \sum_{j=0}^{m-k} \frac{p_{k+j}(x)(y-x)^j}{j!} \right| \leq \frac{K|y-x|^{m+1-k}}{(m+1-k)!}. \quad (7)$$

For $k = m$ this is simply $|p_m(y) - p_m(x)| \leq K|y-x|$. It follows successively from (7), starting at $k = m$, that every p_k is continuous on E . Hence all p_k extend uniquely and continuously to $C = \overline{E}$. Formula (4), viewed after affine normalization to $[0, 1]$, is continuous in two jets at distinct points. Passing to limits in pairs from E therefore preserves $\Gamma_m \leq K$ on C and preserves (7).

Step 2: fill the gaps of C . For each bounded component (α, β) of $\mathbb{R} \setminus C$, choose a minimizer $F_{\alpha\beta}$ for the two endpoint jets. Its cost is at most K . On an unbounded component use the degree- m polynomial determined by the jet at its finite endpoint. On C prescribe $F^{(k)}(x) = p_k(x)$.

For completeness, we verify the pasting at accumulation points. Define G_k to equal p_k on C and $F_{\alpha\beta}^{(k)}$ on each gap. If t in a gap tends to $c \in C$, choose the endpoint e of that gap lying between c and t . Then $e \rightarrow c$, and the Taylor estimate inside the gap, together with continuity of the p_j on C , gives $G_k(t) \rightarrow p_k(c)$. Thus every G_k is continuous. The same decomposition

$$G_k(t) - p_k(c) = (G_k(t) - p_k(e)) + (p_k(e) - p_k(c)),$$

using the first-order part of the Taylor expansions at e and c , yields

$$G_k(t) - p_k(c) = p_{k+1}(c)(t-c) + o(|t-c|) \quad (0 \leq k < m).$$

Therefore $G'_k = G_{k+1}$ also at points of C . The pasted scalar function $F = G_0$ belongs to $C^m(\mathbb{R})$ and has the prescribed jets.

It remains to check the global Lipschitz bound. The function G_m is K -Lipschitz on C by (7) and on every complementary gap by construction. If $s < t$ lie in different pieces, insert the right endpoint of the piece containing s and the left endpoint of the piece containing t . Both inserted points lie in C , and the three Lipschitz bounds add over subintervals whose lengths sum to $t-s$. Hence

$$|G_m(t) - G_m(s)| \leq K(t-s).$$

Thus $F \in C^{m,1}(\mathbb{R})$ and $\text{Lip}(F^{(m)}) \leq K$.

Step 3: exactness. Every global interpolant G restricts on $[x, y]$ to a competitor in (1), so $\Gamma_m(f; x, y) \leq \text{Lip}(G^{(m)})$. Taking the supremum shows that no global interpolant has seminorm below K . The constructed F therefore has equality in (5) and proves (6).

Finally, for $U(z) = J_z^m F$ on D , restriction of F to every pair gives $\Gamma_m(U; D) \leq K$, while the old pairs in E give the reverse inequality. This proves exact minimal extension to every D . $\square$

5 The source axioms P0–P5

Equip $\mathcal{P}_m$ with any Euclidean coefficient metric. The next proposition checks that the functional above fits the framework of [1] on the domain space $\mathbb{R}$.

Proposition 3. *For scalar m -fields on $\mathbb{R}$, the functional $\Phi = \Gamma_m$ satisfies properties (P0)–(P5) of Hirn–Le Gruyer.*

Proof. Symmetry and nonnegativity (P_0) follow from time reversal in (1); pointwise evaluation (P_1) is the definition of $\Gamma_m(f; E)$. The final assertion of Theorem 2 is exactly the minimal-extension property (P_2).

For (P_3), suppose $x < y$ and $[x, y] \subset \text{dom}(f)$. For any $z \in [x, y]$, paste minimizers for the pairs (x, z) and (z, y) . Their jets through order m agree at z , so the pasted function is in $C^{m,1}([x, y])$ and

$$\Gamma_m(f; x, y) \leq \max\{\Gamma_m(f; x, z), \Gamma_m(f; z, y)\}. \quad (8)$$

Taking $z = (1 - t)x + ty$ gives the required monotone curve and, since (8) holds for every t , the required infimum inequality.

For (P_5), if $\Gamma_m(f; E) = K < \infty$, estimate (7) shows first that p_m is Lipschitz and then that every p_k is continuous. Converting centered Taylor coefficients to ordinary polynomial coefficients is continuous in $(x, p_0, \dots, p_m)$, so $f : E \rightarrow \mathcal{P}_m$ is continuous.

For (P_4), normalize a pair of distinct points affinely to 0 and 1. On $\mathbb{R}^{2m+2}$, the resulting two-jet cost is a finite seminorm: homogeneity is clear, and subadditivity follows by adding two interpolants. Every finite seminorm on a finite-dimensional space is continuous. Undoing the affine normalization multiplies the cost by $|x - y|^{-(m+1)}$ and transforms jet coordinates continuously as long as $x \neq y$. Property (P_5) gives $f(z) \rightarrow f(y)$ as $z \rightarrow y$ within the domain; hence

$$\Gamma_m(f; x, z) \longrightarrow \Gamma_m(f; x, y),$$

which is (P_4). □

Corollary 4 (One-dimensional higher-order quasi-AMLEs). *For scalar m -fields on $\mathbb{R}$, the quasi-AMLE existence theorem of Hirn–Le Gruyer applies with $\Phi = \Gamma_m$ for every $m \geq 1$.*

Proof. The real line has the metric hypotheses of [1], and Proposition 3 supplies (P_0)–(P_5). This is precisely the input of their main theorem. □

6 Upgrade attempts, scope, and novelty check

The argument was pushed through five increasingly strong stages: finite ordered data; arbitrary data via closure and gap filling; an explicit L^∞ moment problem; the finite-dimensional dual formula (4); and verification of all six source axioms, yielding the quasi-AMLE corollary. A further attempt to pass to $d \geq 2$ did not close. The line proof uses the total order of $\mathbb{R}$: complementary components have two endpoints and can be filled independently. In higher dimensions, two-point interpolants on different segments cannot be pasted into a single $C^{m,1}$ function, and no analogue of the exact gap decomposition was found. General Whitney extension theory supplies extensions with dimension- and order-dependent constants, but that does not prove the exact pairwise identity required here.

The result is therefore a solved one-dimensional subcase, not a solution of the source’s unrestricted $\mathbb{R}^d$ question. A bounded novelty search was performed on arXiv using the source id and the phrases “ m -fields Γ^m ”, “exact $C^{m,1}$ extension”, “one-dimensional Whitney jets”, “minimal Lipschitz m th derivative”, and “Hermite interpolation L^∞ ”. The closest items found were general Whitney-type and definable extension theorems [2, 3]; they provide existence or bounded extension results, not this exact one-dimensional pairwise functional and the verification of (P_0)–(P_5). No exact match was found. This is a bounded search, so novelty confidence is moderate rather than definitive.

Human review should focus on the closure-and-gap pasting in Theorem 2 and on whether the intended meaning of “correct functional” in the source imposes any convention beyond pairwise

dependence on the two endpoint jets. The functional here is genuinely pair-local and has the explicit formula (4); it is not the circular construction obtained by assigning the same global trace norm to every pair.

References

- [1] M. J. Hirn and E. Y. Le Gruyer, *A general theorem of existence of quasi absolutely minimal Lipschitz extensions*, arXiv:1211.5700v3 (2014).
- [2] P. Shvartsman, *Whitney-type extension theorems for jets generated by Sobolev functions*, arXiv:1607.01660 (2016).
- [3] A. Parusiński and A. Rainer, *Uniform extension of definable $C^{m,\omega}$ -Whitney jets*, arXiv:2306.09156 (2023).